

10.8 (a) From Eq. (8.11), we have  $\vec{E} = \sqrt{\frac{\mu_c \omega}{2\sigma}} (1-i) (\vec{n} \times \vec{H})$ . By definition,

$$Z_s = \sqrt{\frac{\mu_c \omega}{2\sigma}} (1-i). \quad \text{Since } Z_0 = \sqrt{\mu_0/\epsilon_0}, \text{ and } \mu_c \cong \mu_0, \text{ then}$$

$$\frac{Z_s}{Z_0} = \sqrt{\frac{\epsilon_0 \omega}{2\sigma}} (1-i) = \sqrt{\epsilon_0 \mu_c \omega} \cdot \sqrt{\frac{2}{\mu_c \omega \sigma}} \cdot \frac{1}{2} (1-i) = \frac{\omega}{c} \delta \cdot \frac{1}{2} (1-i) = \frac{k\delta}{2} (1-i),$$

$$\text{where } \delta = (2/\mu_c \omega \sigma)^{1/2}.$$

$$(b) \text{ Since } h_0^{(1)}(x) = \frac{e^{ix}}{ix}, \quad h_1^{(1)}(x) = -\frac{e^{ix}}{x} \left(1 + \frac{i}{x}\right), \quad h_2^{(1)}(x) = \frac{ie^{ix}}{x} \left(1 + \frac{3i}{x} - \frac{3}{x^2}\right),$$

$$\begin{aligned} \text{and } \frac{1}{x} \frac{d}{dx} (x h_1^{(1)}(x)) &= \frac{h_0^{(1)}(x)}{2} + \frac{h_1^{(1)}(x)}{2x} - \frac{h_2^{(1)}(x)}{2} \\ &= e^{ix} \left( -\frac{i}{2x} - \frac{1}{2x^2} - \frac{i}{2x^3} - \frac{i}{2x} + \frac{3}{2x^2} + \frac{3i}{2x^3} \right) \\ &= e^{ix} \left( \frac{i}{x^3} + \frac{1}{x^2} - \frac{i}{x} \right) \end{aligned}$$

For  $x \ll 1$ , expand  $e^{ix}$  to third order, we have

$$\begin{aligned} \frac{1}{x} \frac{d}{dx} (x h_1^{(1)}(x)) &= \left( 1 + ix - \frac{x^2}{2} - \frac{i x^3}{6} \right) \left( \frac{i}{x^3} + \frac{1}{x^2} - \frac{i}{x} \right) \\ &= \frac{i}{x^3} - \frac{i}{2x} + \frac{2}{3}. \end{aligned}$$

$$\text{Also, } h_1^{(1)}(x) = -\left( 1 + ix - \frac{x^2}{2} - \frac{i x^3}{6} \right) \left( \frac{1}{x} + \frac{i}{x^2} \right) = -\frac{i}{x^2} - \frac{i}{2} + \frac{x}{3}, \text{ Then, for } ka \ll 1,$$

we can write the denominator as

$$\begin{aligned} h_1^{(1)}(ka) &= i \frac{Z_0}{Z_s} \left[ \frac{1}{x} \frac{d}{dx} (x h_1^{(1)}(x)) \right] \Big|_{x=ka} \\ &= -\frac{i}{k^2 a^2} - \frac{i}{2} + \frac{ka}{3} - i(1-i) \frac{k\delta}{2} \left[ \frac{i}{k^2 a^3} - \frac{i}{2ka} + \frac{2}{3} \right] \\ &= -\frac{i}{k^2 a^2} \left[ 1 + \frac{k^2 a^2}{2} + \frac{i}{3} k^2 a^3 + (1-i) \frac{\delta}{2a} \left( i - \frac{i}{2} k^2 a^2 + \frac{2}{3} k^2 a^3 \right) \right] \\ &= -\frac{i}{k^2 a^2} \left[ \left( 1 + \frac{\delta}{2a} \right) + i \frac{\delta}{2a} \right] - \frac{i}{2} \left[ 1 - (1+i) \frac{\delta}{2a} \right] + \frac{ka}{3} \left[ 1 - (1+i) \frac{\delta}{a} \right] \end{aligned}$$

Similarly, for the numerator, notice that  $h_e^{(2)}(x) = h_e^{(1)}(x)^*$ , then

$$\begin{aligned} h_1^{(2)}(ka) &= i(1-i) \frac{k\delta}{2} \left[ \frac{1}{k} \frac{d}{dx} (x h_1^{(1)}(x)) \right] \Big|_{x=ka} \\ &= \frac{i}{k^2 a^2} + \frac{i}{2} + \frac{ka}{3} - i(1-i) \frac{k\delta}{2} \left[ -\frac{i}{k^3 a^3} + \frac{i}{2ka} + \frac{2}{3} \right] \\ &= \frac{i}{k^2 a^2} \left[ 1 + \frac{1}{2} k^2 a^2 - \frac{i}{3} k^3 a^3 - (1-i) \frac{\delta}{2a} \left[ -i + \frac{i}{2} k^2 a^2 + \frac{2}{3} k^3 a^3 \right] \right] \\ &= \frac{i}{k^2 a^2} \left[ \left(1 + \frac{\delta}{2a}\right) + i \frac{\delta}{2a} \right] + \frac{i}{2} \left[ 1 - (1+i) \frac{\delta}{2a} \right] + \frac{ka}{3} \left[ 1 - (1+i) \frac{\delta}{a} \right]. \end{aligned}$$

$$\begin{aligned} \text{Therefore, } \alpha_{\pm}(1) &= -1 - \frac{h_1^{(2)}(ka) - i \frac{\delta}{2a} \left[ \frac{1}{k} \frac{d}{dx} (x h_1^{(1)}(x)) \right] \Big|_{x=ka}}{h_1^{(1)}(ka) - i \frac{\delta}{2a} \left[ \frac{1}{k} \frac{d}{dx} (x h_1^{(1)}(x)) \right] \Big|_{x=ka}} \\ &= -1 - \frac{\frac{i}{k^2 a^2} \left[ \left(1 + \frac{\delta}{2a}\right) + i \frac{\delta}{2a} \right] + \frac{i}{2} \left[ 1 - (1+i) \frac{\delta}{2a} \right] + \frac{ka}{3} \left[ 1 - (1+i) \frac{\delta}{a} \right]}{-\frac{i}{k^2 a^2} \left[ \left(1 + \frac{\delta}{2a}\right) + i \frac{\delta}{2a} \right] - \frac{i}{2} \left[ 1 - (1+i) \frac{\delta}{2a} \right] + \frac{ka}{3} \left[ 1 - (1+i) \frac{\delta}{a} \right]} \\ &= \frac{\frac{2ka}{3} \left[ \left(1 - \frac{\delta}{a}\right) - i \frac{\delta}{a} \right]}{\frac{i}{k^2 a^2} \left[ \left(1 + \frac{\delta}{2a}\right) + i \frac{\delta}{2a} \right]} = -\frac{2i}{3} (ka)^3 \frac{\left(1 - \frac{\delta}{a}\right) - i \frac{\delta}{a}}{\left(1 + \frac{\delta}{2a}\right) + i \frac{\delta}{2a}} \end{aligned}$$

To calculate  $\beta$ , we have

$$\text{denominator} = -\frac{i}{k^2 a^2} - \frac{i}{2} + \frac{1}{3} ka - i(1+i) \frac{1}{k\delta} \left[ \frac{i}{k^3 a^3} - \frac{i}{2ka} + \frac{2}{3} \right],$$

$$\text{numerator} = \frac{i}{k^2 a^2} + \frac{i}{2} + \frac{1}{3} ka - i(1+i) \frac{1}{k\delta} \left[ -\frac{i}{k^3 a^3} + \frac{i}{2ka} + \frac{2}{3} \right],$$

$$\begin{aligned} \text{and } \beta_{\pm}(1) &= -1 - \frac{\frac{i}{k^2 a^2} + \frac{i}{2} + \frac{1}{3} ka - i(1+i) \frac{1}{k\delta} \left[ -\frac{i}{k^3 a^3} + \frac{i}{2ka} + \frac{2}{3} \right]}{-\frac{i}{k^2 a^2} - \frac{i}{2} + \frac{1}{3} ka - i(1+i) \frac{1}{k\delta} \left[ \frac{i}{k^3 a^3} - \frac{i}{2ka} + \frac{2}{3} \right]} \\ &= \frac{\frac{2}{3} ka + \frac{4}{3} \frac{1-i}{k\delta}}{-\frac{1-i}{k\delta} - \frac{i}{k^3 a^3}} = \frac{4i}{3} (ka)^3 \end{aligned}$$